



HIT140 FOUNDATIONS OF DATA SCIENCE

Assignment-2 (Group Presentation)

Sydney Group 2

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FIFA

FIFA World Cup 2026

Have You ever wondered,
How many tackles it takes to get a yellow card?

What about booking conversion rates of outfield players?,
and Are defenders and midfielders treated differently than
forwards?

The question



Among outfield players at the World Cup 2026, how many yellow cards were received per 10 fouls committed, and is that rate different for defenders and midfielders compared to forwards?

Why not just count cards?

- A player who starts every match and makes a lot of tackles tops the card table anyway.
- Counting mixes up how much a player fouls with how the referee treats those fouls.
- Dividing by fouls takes the workload out and leaves the referee's judgement.

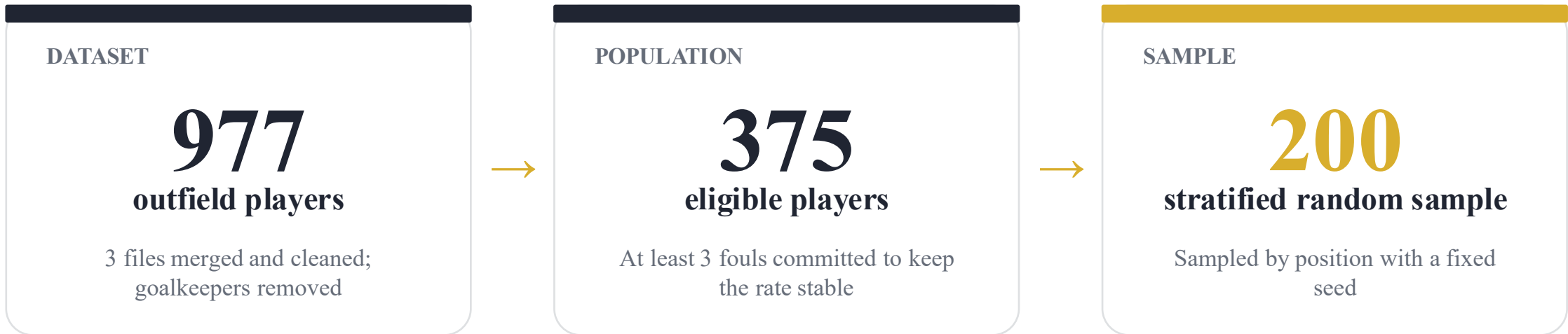
booking rate

$$\frac{\text{yellow cards}}{\text{fouls committed}} \times 10$$

= yellow cards per 10 fouls

Expectation before testing: defenders and midfielders foul closer to their own goal, where a foul is far more likely to be carded.

A cleaned dataset was narrowed to an eligible population, then sampled proportionally by playing position.



QUALITY CHECKS

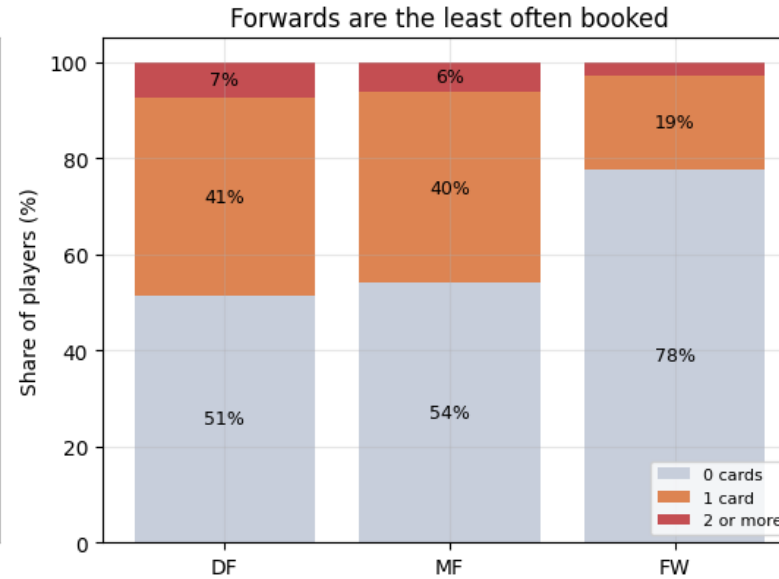
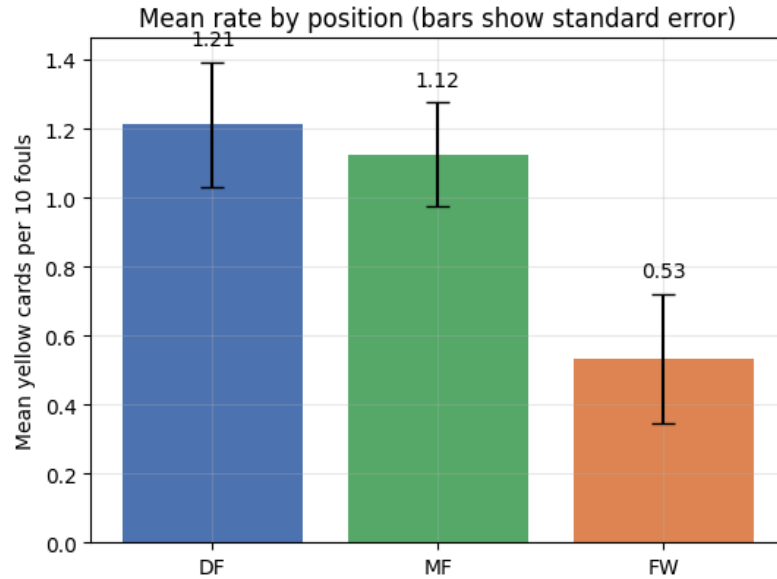
No missing values or duplicate players. No player was credited with more matches than their team played.

WHY STRATIFY?

Position is the variable being compared, and forwards are the smallest group.

Sample mean: 1.047 | Population mean: 1.026 | Position split preserved

What the sample looks like



1.05

mean cards per 10 fouls

0

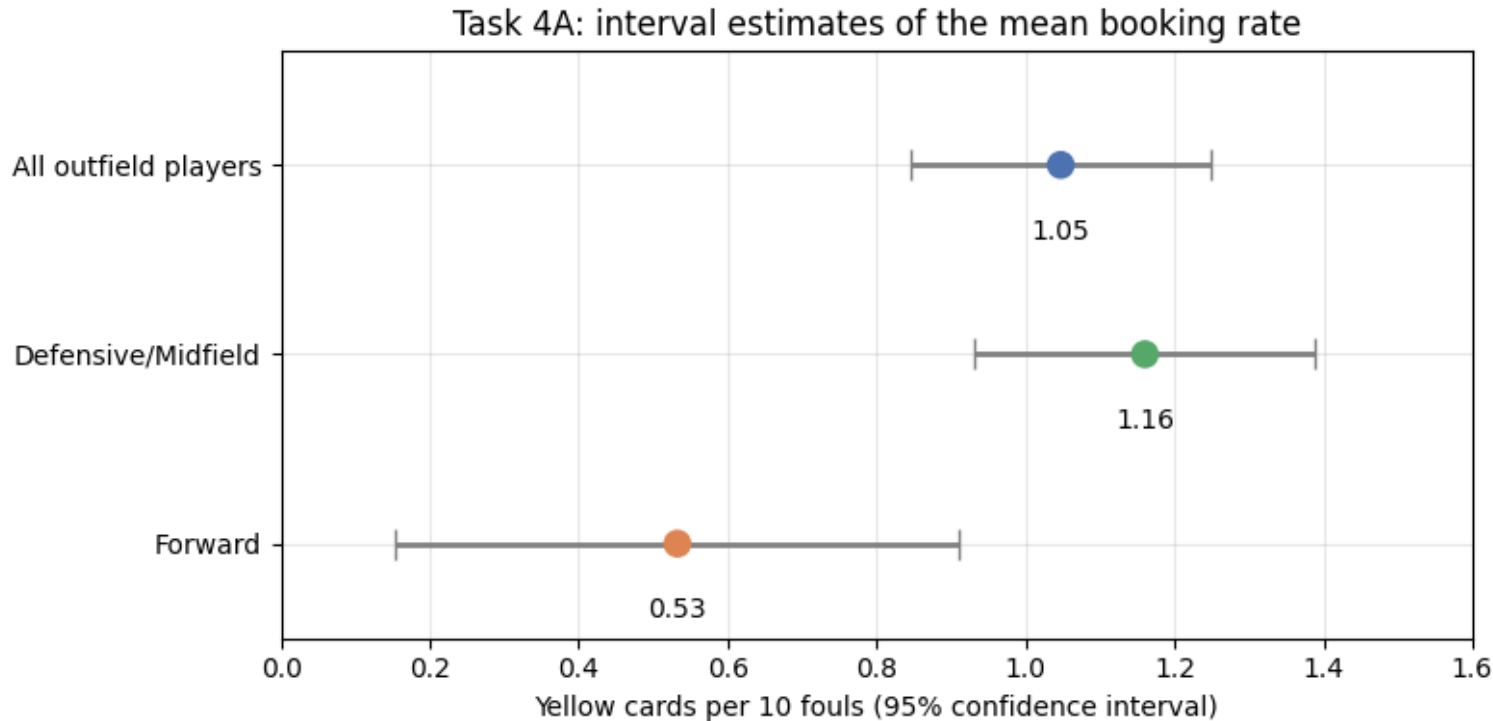
median, most players never booked

57.5%

fouled 3+ times, no card at all

Strongly right skewed. Defenders 1.21 and midfielders 1.12 sit well above forwards 0.53, and 78% of forwards finished with no card at all against 51% of defenders.

Confidence interval



0.85 to 1.25

95% confidence interval for the mean booking rate of all outfield players

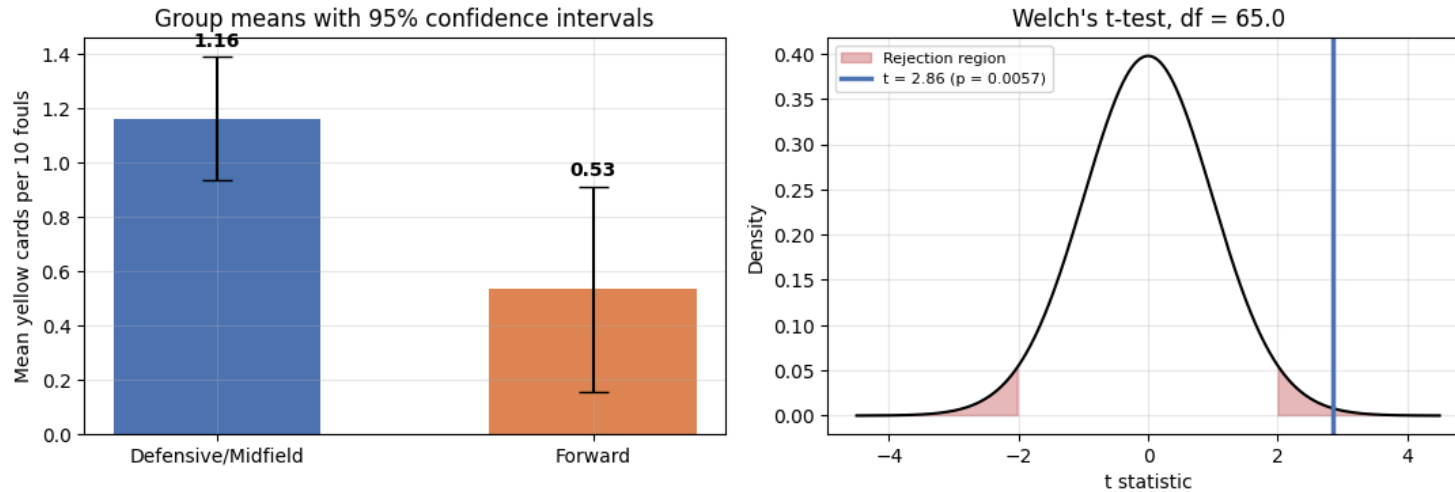
Roughly 1 foul in every 10 was carded.

- The real population mean, 1.026, does fall inside the interval.
- A bootstrap gave almost the same interval, so the skew is not a problem.

Two-sample t-test



Task 4B: comparing booking conversion rates



H0: position makes no difference

H1: the two rates are different

$p = 0.0057$

null hypothesis rejected

- Welch's test used, because Levene's test showed unequal variances ($p = 0.018$).
- Defenders and midfielders 1.16 per 10 fouls against 0.53 for forwards.
- $t = 2.858$, $df = 65.03$, difference CI $[0.19, 1.07]$ excludes zero, Cohen's $d = 0.44$.

What it means



1. About 1 foul in 10 was carded

Roughly 8 to 12 percent of fouls at the tournament ended in a yellow card, and most players who fouled were never booked at all.

2. Where a player fouls beats how much

Defenders and midfielders are carded twice as often per foul as forwards, which fits Law 12 on stopping promising attacks.

3. Counting cards would have missed this

Raw counts mix workload with referee judgement. The gap survives once fouls are used as the denominator.

Limitations: tournament totals only, so foul location, severity and timing are not observable. Position is a stand in for all of that.



Thank You!